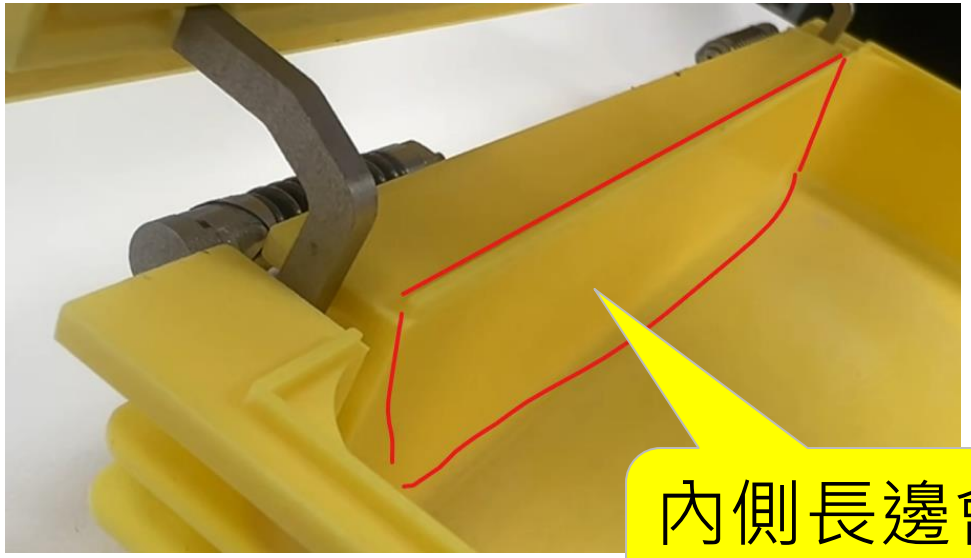
# Profolio

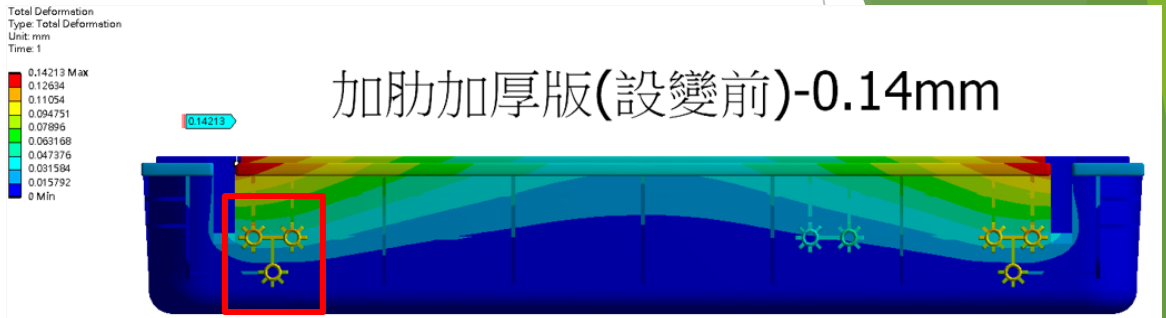
張凱鈞 CAE結構分析資深工程師

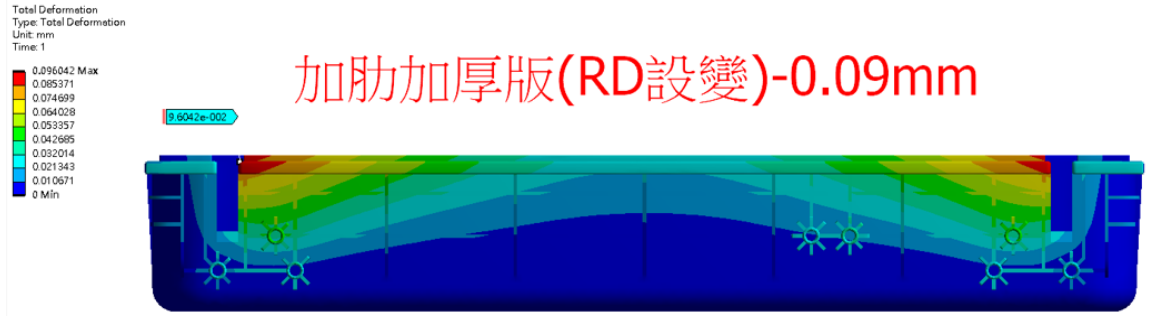
2026.10.16

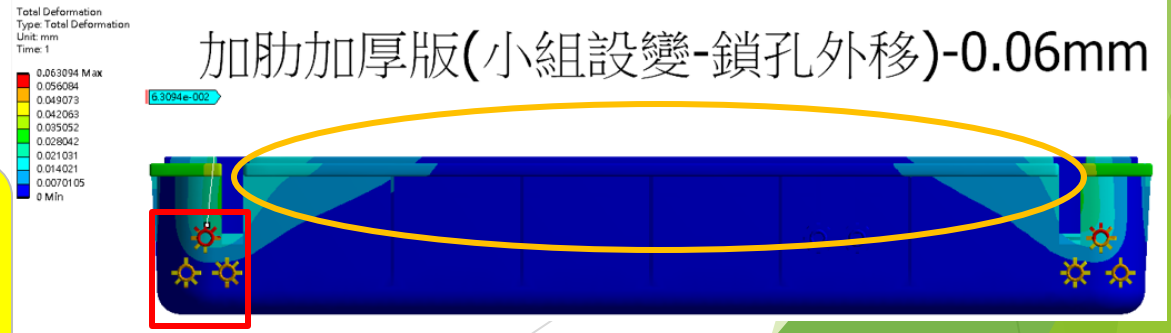
# 提供解決方案-靜力結構分析



內側長邊會前後晃動



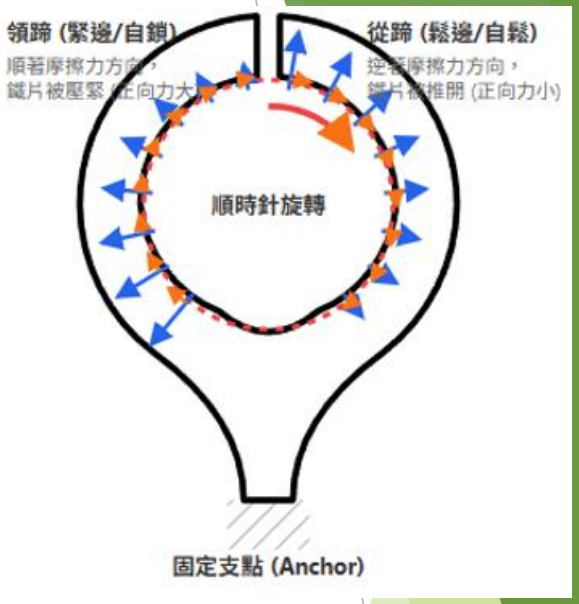
# DOE-扭力值分析

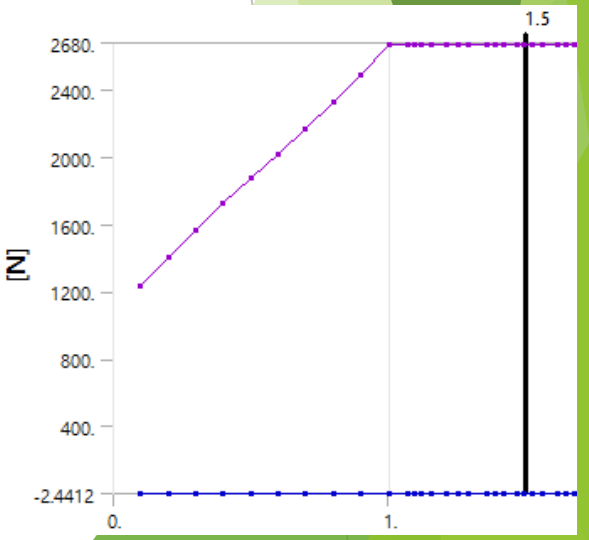
實際扭力=摩擦係數 x 分析反作用力 x 心軸半徑
T=μ x N x r

抓摩擦係數:
μ=(實際扭力*98)/(心軸半徑*分析反作用力)
μ=(2.6*98)/(1.2275*2672.6)=0.077

**運用:**
**設變幾何，分析取得反作用力，可快速取得目標扭力值的設計，減少開模費用至少20%**

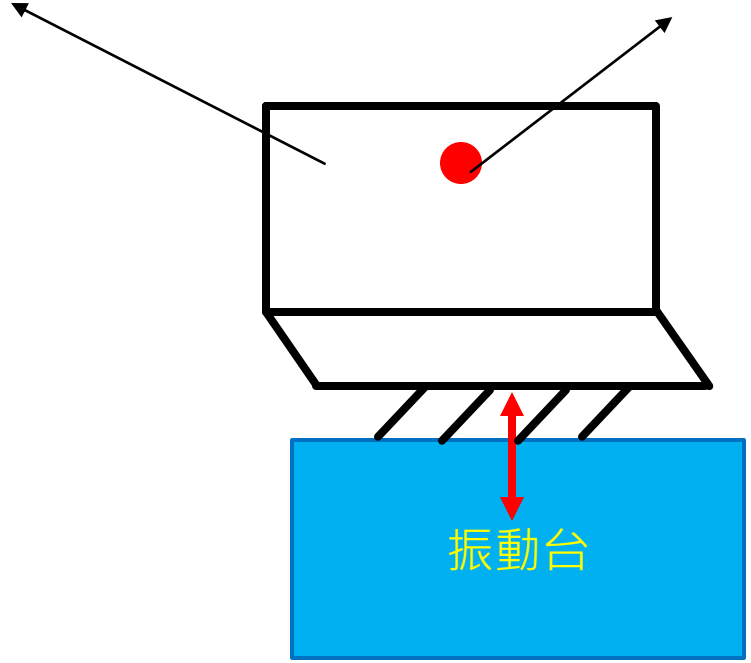
| 實測扭力 (kgf-cm) | 摩擦係數 | 目標扭力值 (kgf-cm) |
|---|---|---|
| 2.6 | ? | 4.0 |



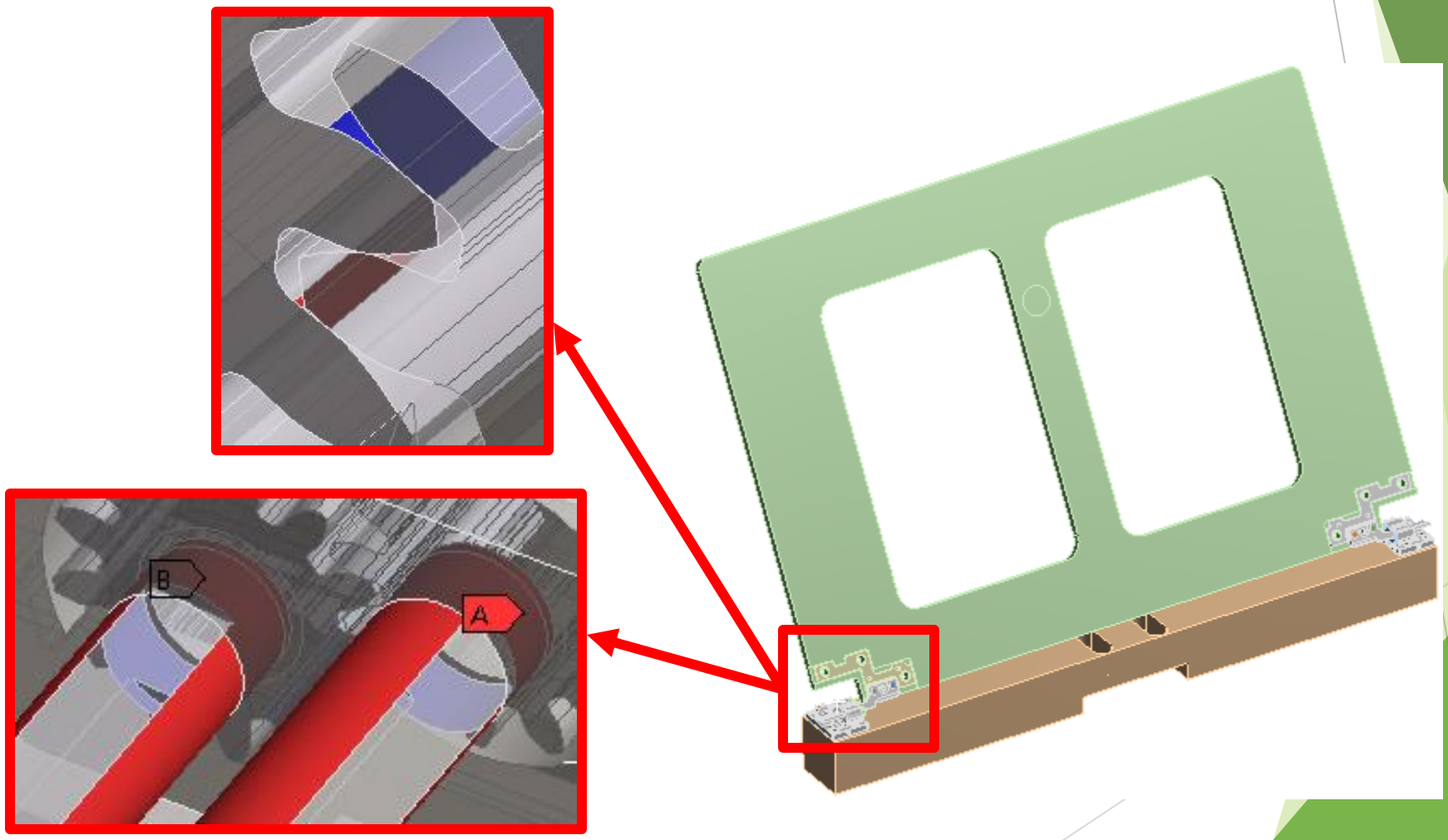
# 實測與分析數據對齊
## -簡諧響應分析(Harmonic Response)

提供的資料：面板重量、固定條件、掃頻範圍、加速度、測試數據



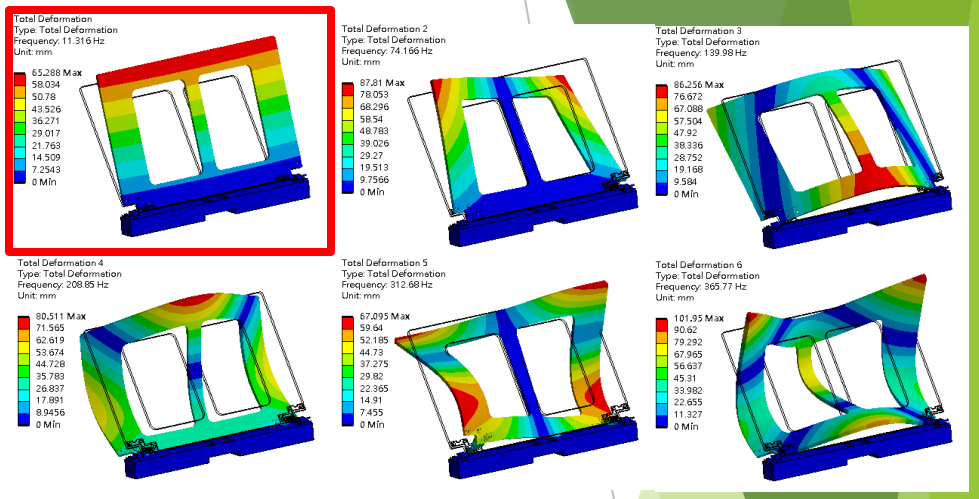

| 共振頻率(Hz) | 振幅(in) |
|---|---|
| 11.17 | 0.06 |

# 分析接觸設定-評估合理的接觸設定



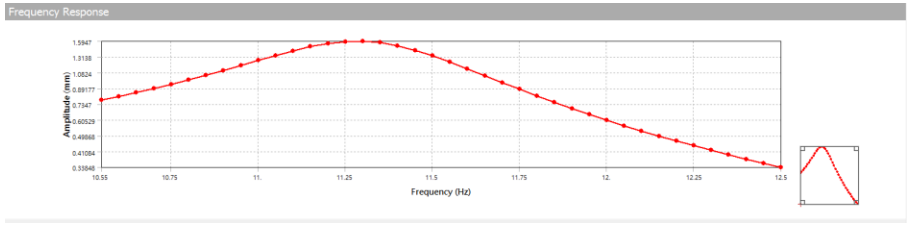
# Modal分析

判斷合理的模態振型和共振頻率(11.3Hz)




# Harmonic Response

調整阻尼比找到對應的振幅




| 結果比較 | | |
|---|---|---|
| | 實驗值 | 分析值 |
| 共振頻率 (Hz) | 11.17 | 11.3 |
| 振幅(in) | 0.06 | 0.062 |

# 分析報告-應力值自動鍵入小工具，從分到秒

## 使用工具：Claude AI 免費版 進行Vibe Coding




從分到秒

analysis report-2026-07-16.pptx



# 謝謝聆聽!